

Robustness of Phishing Detection Features Across Multiple Datasets

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1 Introduction

Motivation & Problem Statement Phishing attacks remain one of the most common and accessible forms of cybercrime, affecting individuals and organizations alike. Unlike other attack vectors that exploit software vulnerabilities, phishing relies on social engineering—mimicking legitimate websites to deceive victims into surrendering sensitive information or downloading malware. Traditionally, defenses have relied on blacklists maintained by the community to block connections to known malicious domains. While moderately effective, blacklists are reactive; they struggle to keep pace with the rapid generation of zero-day phishing sites. This challenge is exacerbated by the rise of Generative AI and Large Language Models (LLMs), which allow attackers to automate the creation of convincing phishing pages and refine their forgery techniques at unprecedented speeds. (Gupta, Akiri, Aryal, Parker, & Praharaaj, 2023)

The Gap (Generalization Issues): While AI offers promising countermeasures through machine learning-based detection systems, these solutions face a critical limitation: the generalization gap. High-performance models often suffer from a significant decline in accuracy when deployed in environments that differ from their training data. This lack of robustness is particularly dangerous in cybersecurity, where the threat landscape constantly evolves. Furthermore, many high-end models operate as "black boxes," making it difficult to decipher why a model fails or how it reaches a decision. While the models themselves are complex, the datasets they are trained on offer a tangible starting point for investigation. A detection model is only as robust as the features it learns, yet little is known about which feature families remain predictive across diverse data sources.

Contributions To address this uncertainty, this supervised learning experiment investigates the robustness of phishing detection features across multiple datasets. We aim to quantify how specific features affect model performance and stability when transferred between different data environments. Our goal is to create a reliable feature ranking for phishing website detection that holds true across multiple environments, moving beyond simple accuracy metrics to understand the underlying stability of the learned patterns.

Research Questions Consequently, our experiment aims to answer the following research questions:

1. How consistent are feature importances (overlapping features) across diverse phishing datasets?
2. Which features degrade or invert their predictive behavior when applied cross-dataset?
3. Does merging data sources improve generalization?
4. Which feature families (lexical, host, content) are most domain-stable?

2 Related Works

While numerous studies have demonstrated high accuracy within isolated environments, recent scholarships have increasingly questioned the reliability of these results when applied to the open web. A primary critique is that many proposed systems are "dataset dependent," meaning their high performance does not generalize to new or evolving phishing tactics.

Benchmark Datasets and Feature Evaluation: (Hannousse & Yahiouche, 2021a) To address the lack of reproducible standards, Hannousse and Yahiouche developed a comprehensive framework for building benchmark datasets, directly responding to the "impracticable replication" of prior studies. They constructed a dataset of 11,430 legitimate and phishing URLs (which is utilized in this project), extracting 87 distinct features across three categories: URL, HTML content, and external services.

In their experimental analysis, they found that Random Forest was the most predictive classifier, achieving an accuracy of 96.61% when using hybrid features. Their findings suggested that external service-based features (e.g., WHOIS, PageRank) were the most discriminative, whereas content-based features were often less distinguishing and computationally expensive for runtime detection. This distinction is vital for our study, as we aim to verify if these "discriminative" external features remain robust when tested across entirely different datasets.

Cross-Dataset Generalization: (Mia, Derakhshan, & Pritom, 2024) Mia et al. investigated the trustworthiness of features when transferred between diverse datasets. In their study, the authors trained models on one source and tested them on others, utilizing Explainable AI (SHAP) to visualize feature importance. They discovered that feature rankings were highly inconsistent; a feature considered top in one dataset often degraded in predictive power in another. Their work highlighted the generalization gap, the decline in performance when models face environments different from their training data.

Hannousse and Yahiouche’s analysis was primarily "within-dataset" (using cross-validation on a single benchmark), while Mia et al. focused heavily on lexical URL features, neglecting the rich DOM and content features available in modern datasets. This project bridges these two approaches. We utilize the rich feature set defined by Hannousse et al. (URL, Content, External) but apply the cross-dataset testing methodology of Mia et al. Furthermore, for our work:

- **Quantifying Robustness:** Moving beyond visual SHAP plots to calculate a specific Robustness Score (R) for each feature.
- **Expanding Scope:** Testing if "less distinguishing" content features (as labeled by Hannousse) perhaps offer better stability across domains than the highly discriminative but potentially volatile external features.

3 Methodology

To evaluate the stability of phishing detection features across diverse environments, we designed a multi-stage experimental framework implemented in Python and JupyterNotebook. This framework involves the mapping of three distinct datasets, the derivation of missing features from raw identifiers, and the application of Explainable AI (XAI) to quantify feature robustness.

3.1 Data Sources

We utilized three datasets to ensure temporal and structural diversity.

- **Dataset A:** A balanced dataset containing 10,000 instances. It provides 48 pre-extracted features with a specific focus on HTML and JavaScript behaviors. We convert arff file to csv using Pulipulichen’s online tool.(Tan, 2018)(Pulipulichen, n.d.)
- **Dataset B:** A benchmark dataset comprising 11,430 instances. This dataset is unique in that it provided both pre-extracted features and the raw URLs, allowing us to perform hybrid feature extraction.(Hannousse & Yahiouche, 2021b)
- **Dataset C:** A larger repository from which we utilized a subset of 27,887 raw (HTML + URLs) instances. We extract features using a python script with BeautifulSoup and NLP. We ensure to mix variety from both dataset A, B features (for common feature experiment) and mix with our own features (for dataset own identity). This dataset required extensive preprocessing to align its raw content-based counts with the boolean flags used in other datasets.(Ariyadasa, Fernando, & Fernando, 2021)

3.2 Feature mapping and Preprocessing

A primary challenge in cross-dataset analytics is feature mismatch. To address this, we implemented a five-stage preprocessing pipeline:

1. **Standardization & Mapping:** Column names were normalized to lowercase with special characters removed. We constructed a `COLUMN_MAPPING` dictionary to map diverse nomenclatures (e.g., `nb_dots`, `numdots`) to a unified schema (e.g., `num_dots`).
2. **Lexical Derivation (Dataset B):** We utilized the `urllib.parse` library to mathematically derive missing lexical features. This included calculating `path_depth`, `query_length`, and `num_numeric_chars` directly from the source strings to ensure parity with Dataset A and C.

3. **Semantic Alignment:** We corrected semantic divergences between datasets. For example, continuous count features in Dataset C were converted into boolean flags to match the binary nature of Datasets A and B. Inverse logic was applied to inconsistent flags.
4. **Imputation and Cleaning:** Columns with greater than 50% missing values were dropped. Remaining missing values were imputed using a dual strategy: binary columns were filled with 0, while continuous numerical columns were imputed using the median to preserve distribution properties.
5. **Scaling:** All features were normalized using Z-score standardization (StandardScaler) independently for each dataset to prevent data leakage.

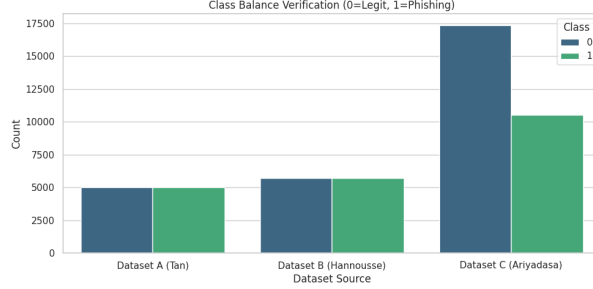


Figure 1: Legit/Phishing distribution after pre-processed

Overlap Feature: We identified an intersection of 23 common features present across all three datasets. These features, spanning lexical, host-based, and content-based, form the basis of our robustness analysis.

3.3 Experimental Protocols and Classifiers

To assess generalization, we implemented three distinct training protocols using 5-fold cross-validation with an 80/20 train-test split:

- **Baseline (Within-Dataset):** Models were trained and tested within the same dataset to establish a baseline.
- **Generalization Assessment (Cross-Dataset):** Models trained on one dataset (e.g., A) were tested directly on another (e.g., B).
- **Merged Data Analysis:** Datasets were merged to evaluate if diverse training sources improve stability.

We employed three classifiers: Logistic Regression (linear baseline), Random Forest (ensemble tree-based), and FT-Transformer (deep learning).

3.4 Robustness Metrics

We quantified feature stability using two custom metrics. First, the Generalization Gap (ΔAUC) measures performance degradation (Equation 1). Second, to quantify the relationship between a feature’s predictive power and its volatility, we employed Spearman’s Rank Correlation Coefficient (ρ) between feature mean importance and std. Third, the Robustness Score (R) ranks features based on the stability of their SHAP importance values across datasets (Equation 2):

$$\Delta AUC = AUC_{in_dataset} - AUC_{cross_dataset} \quad (1)$$

$$R = \frac{\mu_{importance}}{1 + \sigma_{importance}} \quad (2)$$

A high R value indicates a feature that is consistently predictive regardless of the training environment.

4 Result

4.1 Protocol 1: Within-Dataset Performance

Table 1: Comparison of different models with variety datasets using their **full features**

Dataset	Model	Accuracy	F1 Score	ROC AUC	PR AUC	Precision	Recall
A	Logistic Regression	0.95	0.95	0.99	0.99	0.94	0.96
	Random Forest	0.99	0.99	1.00	1.00	0.99	0.99
	FT-Transformer	0.98	0.98	1.00	1.00	0.97	0.98
B	Logistic Regression	0.94	0.94	0.98	0.98	0.94	0.94
	Random Forest	0.96	0.96	0.99	0.99	0.96	0.96
	FT-Transformer	0.96	0.96	0.99	0.99	0.96	0.95
C	Logistic Regression	0.92	0.90	0.97	0.96	0.90	0.90
	Random Forest	0.97	0.96	0.99	0.99	0.96	0.95
	FT-Transformer	0.96	0.94	0.99	0.99	0.93	0.95

Table 2: Comparison of different models with variety datasets using only **common features**

Dataset	Model	Accuracy	F1 Score	ROC AUC	PR AUC	Precision	Recall
A	Logistic Regression	0.86	0.86	0.95	0.95	0.87	0.86
	Random Forest	0.98	0.98	1.00	1.00	0.98	0.99
	FT-Transformer	0.97	0.97	0.99	0.99	0.97	0.98
B	Logistic Regression	0.79	0.78	0.87	0.87	0.80	0.76
	Random Forest	0.89	0.89	0.95	0.96	0.90	0.88
	FT-Transformer	0.86	0.86	0.94	0.94	0.90	0.82
C	Logistic Regression	0.86	0.81	0.93	0.90	0.84	0.78
	Random Forest	0.95	0.93	0.99	0.98	0.93	0.93
	FT-Transformer	0.92	0.89	0.98	0.97	0.92	0.87

4.2 Protocol 2: Cross-Dataset Generalization

Table 3: Comparison of cross-dataset performance of different models

Train $\rightarrow$ Test	Model	Acc	F1	ROC AUC	PR AUC	Prec	Rec	Gen. Gap
A $\rightarrow$ B	Logistic Reg.	0.69	0.66	0.76	0.80	0.74	0.60	0.19
	Random Forest	0.60	0.44	0.70	0.67	0.74	0.32	0.30
	FT-Transformer	0.65	0.54	0.75	0.72	0.80	0.41	0.25
A $\rightarrow$ C	Logistic Reg.	0.73	0.67	0.80	0.76	0.62	0.73	0.15
	Random Forest	0.79	0.67	0.87	0.79	0.81	0.57	0.13
	FT-Transformer	0.81	0.76	0.88	0.81	0.72	0.80	0.12
B $\rightarrow$ A	Logistic Reg.	0.70	0.71	0.77	0.74	0.68	0.74	0.10
	Random Forest	0.67	0.66	0.73	0.71	0.68	0.64	0.23
	FT-Transformer	0.73	0.75	0.80	0.75	0.71	0.79	0.14
B $\rightarrow$ C	Logistic Reg.	0.70	0.64	0.76	0.72	0.58	0.72	0.11
	Random Forest	0.73	0.67	0.81	0.73	0.62	0.73	0.15
	FT-Transformer	0.71	0.66	0.80	0.73	0.60	0.72	0.14
C $\rightarrow$ A	Logistic Reg.	0.70	0.63	0.80	0.80	0.80	0.52	0.13
	Random Forest	0.68	0.54	0.83	0.84	0.94	0.38	0.16
	FT-Transformer	0.78	0.74	0.91	0.89	0.88	0.64	0.07
C $\rightarrow$ B	Logistic Reg.	0.69	0.63	0.73	0.78	0.79	0.52	0.20
	Random Forest	0.67	0.61	0.74	0.75	0.76	0.51	0.24
	FT-Transformer	0.72	0.69	0.80	0.81	0.75	0.64	0.18

4.3 Protocol 3: Mixed Dataset Training

Table 4: Compare model performance with mixed training dataset

Train $\rightarrow$ Test	Model	Acc	F1	ROC AUC	PR AUC	Prec	Rec
A+B $\rightarrow$ C	Logistic Reg.	0.75	0.70	0.83	0.79	0.64	0.77
	Random Forest	0.82	0.74	0.88	0.81	0.79	0.71
	FT-Transformer	0.79	0.75	0.89	0.83	0.67	0.85
A+C $\rightarrow$ B	Logistic Reg.	0.68	0.62	0.74	0.77	0.77	0.52
	Random Forest	0.68	0.61	0.77	0.76	0.79	0.50
	FT-Transformer	0.69	0.70	0.78	0.79	0.68	0.72
B+C $\rightarrow$ A	Logistic Reg.	0.74	0.70	0.83	0.83	0.82	0.61
	Random Forest	0.69	0.58	0.83	0.84	0.92	0.42
	FT-Transformer	0.74	0.68	0.89	0.88	0.90	0.55

Table 5: Consistency of Feature Importance Rankings Across Domains (Spearman Correlation mean feature’s importance vs standard deviation)

Model	Spearman Correlation (ρ)
Logistic Regression	0.9140
Random Forest	0.9061
FT-Transformer	0.8984

Table 6: Feature Robustness Ranking

Model	Top Feature	Mean Imp.	Std. Dev	Robustness (R)
Logistic Regression	url.length	0.119	0.106	0.108
	path.length	0.121	0.163	0.104
	ratio_external_links	0.054	0.038	0.052
	num_dots	0.040	0.032	0.039
	hostname.length	0.040	0.022	0.039
Random Forest	ratio_external_links	0.133	0.043	0.127
	ratio_null_links	0.040	0.037	0.039
	num_numeric_chars	0.040	0.035	0.038
	path.length	0.037	0.032	0.036
	path.depth	0.033	0.039	0.032
FT-Transformer	ratio_external_links	0.087	0.055	0.083
	path.length	0.051	0.063	0.048
	path.depth	0.045	0.043	0.043
	num_numeric_chars	0.040	0.023	0.039
	ratio_null_links	0.033	0.036	0.032

5 Discussion

Baseline experiment (Tables 1 and 2) confirm that all three models—Logistic Regression, Random Forest, and FT-Transformer—can effectively learn phishing patterns within a closed environment. Random Forest consistently dominated the baseline metrics, achieving near-perfect scores when trained on full features. However, a critical insight emerges when comparing Table 1 (Full Features) to Table 2 (Common Features). We observed a distinct performance penalty when restricting models to the 23 overlapping features. This confirms that many highly predictive features are dataset-specific. While excluding these features is necessary for cross-dataset compatibility, it highlights that "universal" phishing detection is inherently more difficult than detecting phishing within a single source.

The generalization gap is quantified in Table 3. Despite high baseline performance, all models suffered precipitous drops in accuracy when tested on unseen datasets, validating our hypothesis regarding the fragility of single-dataset models. Interestingly, Random Forest, which was the strongest performer in the baseline, exhibited the most severe generalization gaps. This suggests that Random Forest’s decision trees likely overfit to dataset-specific artifacts—learning the “noise” of the collection method rather than the robust characteristics of phishing attacks. In several transfer scenarios, the FT-Transformer demonstrated a significantly lower generalization gap compared to Random Forest. This indicates that the attention mechanisms in transformer architectures may be better suited for learning abstract, transferable feature representations than a tree classifier.

Mitigation via data diversity Our hypothesis that “merging data sources improves generalization” is strongly supported by the results in Table 4. The trend show all perform better when mix training set compare to their single dataset protocol. This suggests that exposing models to multiple training data forces them to abandon dataset-specific shortcuts (which don’t work across mixed sources) and focus on the truly invariant features of phishing websites.

The feature stability-utility trade-off Our analysis reveals a critical “stability-utility trade-off” inherent in phishing detection. As shown in Table 5, the Spearman rank correlation (ρ) between Mean Feature Importance and Standard Deviation is consistently high across all models. This strong positive correlation indicates that the most predictive features (those with high mean importance) are also the most volatile (high standard deviation) when transferred between datasets. In other words, the features that “do the heavy lifting” in prediction are often dataset-specific and not hold up in new environment. Conversely, the “stable” features tend to be those that are consistently unimportant. This statistically confirms our hypothesis that feature rankings are inherently inconsistent when transferred, showing the difficulty of identifying features that are both highly predictive and universally stable

Feature ranking While Logistic Regression prioritized `url_length`, both the Random Forest and FT-Transformer identified `ratio_external_links` as the single most robust feature, achieving Robustness Scores (R) of 0.127 and 0.083 respectively. Since this feature also ranked in the top three for Logistic Regression, we conclude that `ratio_external_links` (a content-based feature) is the most trustworthy individual predictor for cross-domain detection. Although the single most robust feature was content-based, the aggregate top rankings are dominated by the Lexical family (e.g., `url_length`, `path_length`, `num_numeric_chars`, `num_dots`). This suggests that while content analysis is valuable, the structural composition of the URL remains the most consistent signal family across different eras and collection methods.

6 Conclusion

This project investigated the challenge of model generalization in phishing detection (website specifically), moving further the usual metrics to quantify the robustness of detection features. Our experiments revealed a significant generalization gap, where even models achieving near-perfect accuracy in closed environment can suffered from performance drop for new data reveal many top feature are dataset specific rather than indicate phishing. Furthermore, we established that data diversity is a prerequisite for robustness forcing model to avoid learning dataset specific pattern. Through our Robustness Score (R) analysis, we identified that while predictive power and stability often conflict, specific features such as `ratio_external_links` and structural lexical attributes maintain consistency across domains.

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